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9702/32

October/November 2008

2 hours

Additional Materials: As listed in the Confidential Instructions.

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

You will be allowed to work with the apparatus for a maximum of one hour for each question.

You are expected to record all your observations as soon as these observations are made, and to plan the presentation of the records so that it is not necessary to make a fair copy of them. The working of the answers is to be handed in.

Additional answer paper and graph paper should be submitted only if it becomes necessary to do so.

You are reminded of the need for good English and clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

All questions in this paper carry equal marks.

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1	
2	
Total	

This document consists of **8** printed pages and **4** blank pages.

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- 1** In this experiment you will investigate the equilibrium position of a pivoted wooden strip as the turning load on it is varied.

(a) The equipment is set up as shown in Fig. 1.1.

- (i)** Add sand to the plastic container until the angle θ is between 5° and 10° .

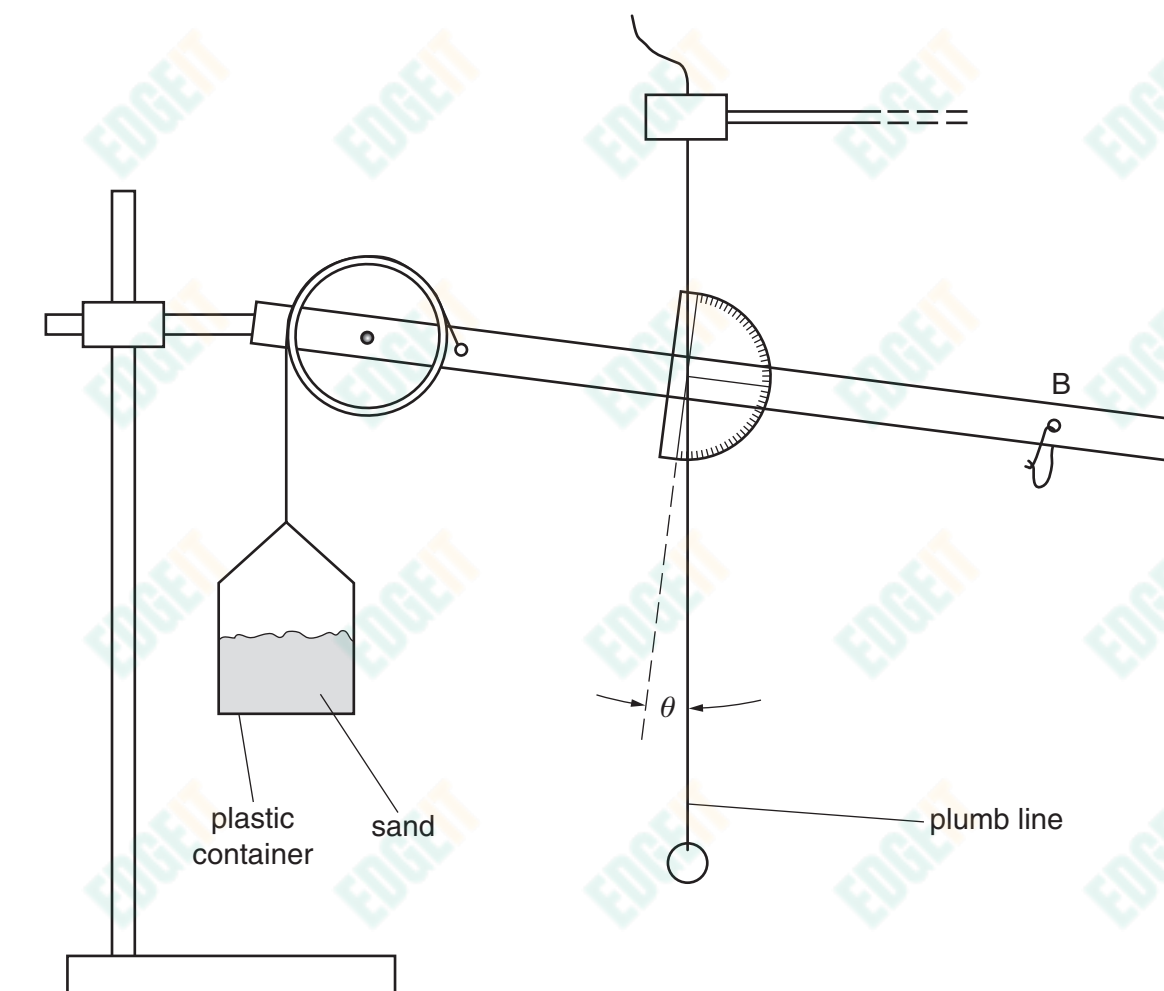


Fig. 1.1

- (ii)** Measure and record the value of θ .

$\theta = \dots\dots\dots^\circ$

- (b) Add n paper clips to the string loop at B and repeat (a)(ii) until you have six sets of values of n and θ , with θ in the range 5° to 60° . You may bend the paper clips if you wish.

Include values of $\frac{1}{\cos\theta}$ in your table of results.

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- (c) Plot a graph of $\frac{1}{\cos\theta}$ on the y -axis against n on the x -axis and draw the line of best fit.
- (d) Determine the gradient and y -intercept of the line.

gradient =

y -intercept =

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- (e) The formula that relates θ and n is

$$\frac{1}{\cos \theta} = 2kmn + kM$$

where k is a constant, m is the mass of one paperclip and M is the mass of the wooden strip and protractor.

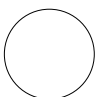
The value of m is written on a card.

Use your answers from (d) to determine values for k and M .

$$k = \dots\dots\dots \text{g}^{-1}$$

$$M = \dots\dots\dots \text{g}$$

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Please turn over for Question 2.

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- 2** In this experiment you will investigate how the period of oscillation of a metal strip varies with its length.

- (a)** You are provided with a metal strip with a mass fixed to it at one end.

Clamp the strip between the two blocks of wood, so that the length l between the blocks and the centre of the mass is about 20 cm as shown in Fig. 2.1.

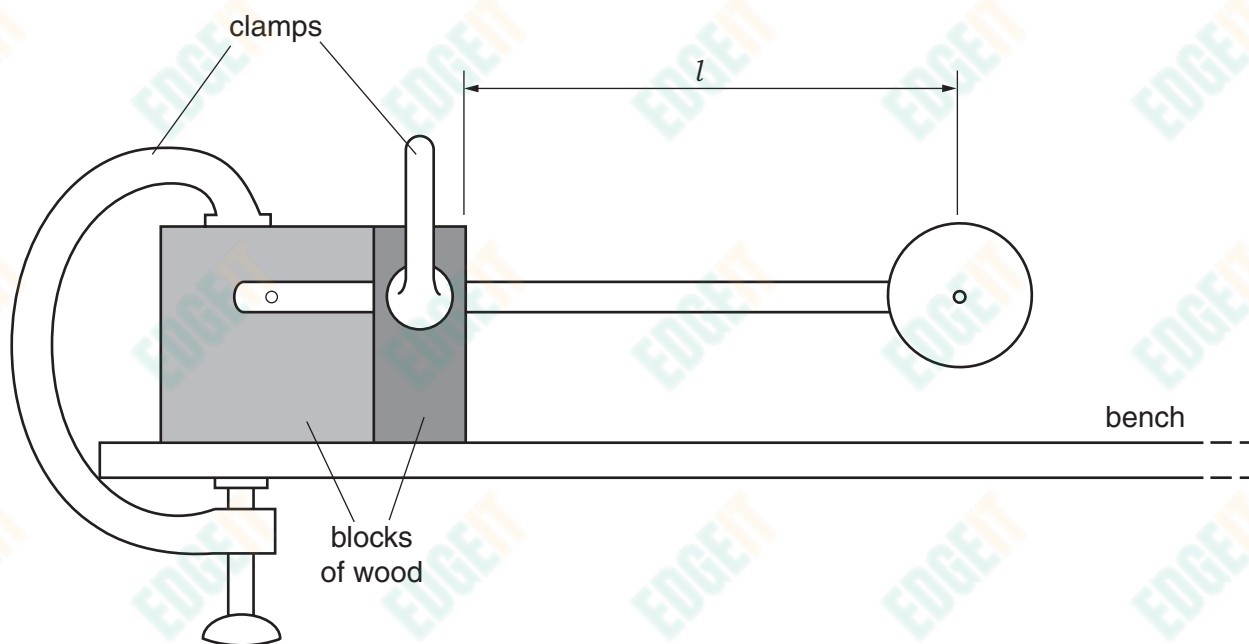


Fig. 2.1

- (b) (i)** Measure the length l .

$l =$ cm

- (ii)** Estimate the percentage uncertainty in your value of l .

percentage uncertainty =%

- (iii)** Calculate the value of l^3 .

$l^3 =$ cm³

- (iv)** Justify the number of significant figures you have given for l^3 .

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- (c) Pull the mass about 2 cm to one side and release it so that it oscillates.

Take measurements to determine the period of oscillation T .

$T = \dots\dots\dots$ s

- (d) Adjust the length l to about 15 cm.

Measure this new length l and repeat (b)(iii) and (c).

$l = \dots\dots\dots$ cm

$l^3 = \dots\dots\dots$ cm³

$T = \dots\dots\dots$ s

- (e) Explain whether your results support the idea that T^2 is proportional to l^3 .

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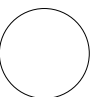
- (f) (i) State four sources of error or limitations of the procedure in this experiment.

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- (ii) Suggest four improvements that could be made to the experiment. You may suggest the use of other apparatus or different procedures.

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